INCH POUND
MIL-PRF-49166/1A(CR)
27 August 1997
SUPERSEDING
MIL-S-49166/1(CR)
8 April 1982

PERFORMANCE SPECIFICATION SHEET
MECHANICAL SCANNER, MX-9872(V)1/UA

This performance specification sheet is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)1/UA Scanner shall consist of this document and the General Specification MIL-PRF-49166.

1. SCOPE

1.1 Scope. This performance specification covers the Mechanical Scanner, MX-9872 (V) 1/UA and augments MIL-PRF-49166. The MX-9872(V)1/UA is used on the AN/ASQ-170, Target Acquisition and Designation Sight and the AN/ASQ-11, Pilot Night Vision Sensor (TADS/PNVS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-49166	- Mechanical Scanner, MX-9872(V)/UA; General Specification for
MIL-PRF-49177	- Scan-Interlace, PL-1408/UA

(Unless otherwise indicated, copies of the above specification is available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation(see 6.2). The drawings below are identifed to ensure dimensions, tolerences and interface requirements are understood and that items built IAW this specification will properly and effectively interface with existing hardware.

DRAWINGS

USA CECOM

SM-D-657713	- Cover, Scanner Housing
SM-D-657620	- Housing, Scanner
SM-D-807690	Scanner
DL-SM-B-773232	Interface Assembly

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2.d), the contractor shall furnish first article units for inspection (see 4.2).

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3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The MX-9872(V)1/UA Scanner, herein known as the MX-9872(V) 1/UA, shall consist of the MX-9872(V)/UA Scanner adjusted for 60 Hz operation.

3.4 Construction. The MX-9872(V)1/UA, SM-D-807690-1 shall meet the form, fit, function, and interface requirements of this specification.

3.4.1 Weight. Weight of the MX-9872(V)1/UA shall be 1.45 pounds maximum as weighed with a scale having an accuracy of 1% or better.

3.4.2 Dimensions. The dimensions and dimensional tolerances of the MX-9872(V)1/UA shall be in accordance with Drawings SM-D-657713 and SM-D-657620 to ensure that items built IAW this specification properly interface with the existing hardware.

3.5 Performance characteristics. The characteristics specified herein shall apply as constructed (3.4). Operation shall be at 60 Hz using the Scan-Interlace, PL-1408/UA

3.5.1 Scan angle and frequency.

3.5.1.1 Variable scan frequency. Applicable for first article test only. See MIL-PRF-49166.

3.5.1.2 Specific scan frequency (30 Hz). Scan frequency of the MX-9872(V)1/UA shall be adjusted to 60.0, ±0.5 Hz. Scan frequency drift shall not be greater than ±6.0 Hz from the frequency at ambient setting when exposed to high and low temperature environmental conditions (see MIL-PRF-49166).

3.5.1.3 Synchronous scan frequency. The MX-9872(V)1/UA shall synchronize with a stable external synchronization source, such as the TV camera vertical sync pulse. Scan frequency shall be stable within 1 Hz of the input sync pulse frequency.

3.5.1.4 Scan angle. The scan angle shall be adjusted to 13.0, ±0.3 degrees.

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3.5.2 Interlace. The MX-9872(V)1/UA shall operate in the 2:1 interlace mode. The interlace switching time shown in Figure 1 of MIL-PRF-49166 shall not be greater than 2.5 milliseconds (ms).

3.5.3 Scan direction. The MX-9872(V)1/UA shall show the waveform shapes and phase relationships in Figure 1 while operating in the 2:1 (side) or the 2:1 (rear) interlace mode.

3.5.4 Power consumption. Power required to operate the MX-9872(V)1/UA shall not exceed 7.5 watts.

3.5.5 Scan efficiency. The MX-9872(V)1/UA scan efficiency shall be a minimum of 70 percent.

3.5.6 Scan jitter in azimuth. Jitter time shall be 12.5 microseconds maximum.

3.6 Workmanship. All parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall be performed on a specified number of scanners when a first article inspection is required (see 3.1). This inspection shall include the examination of paragraph 4.4, and the tests of 4.5.1.2 and 4.5.1.3, 4.5.3 included herein; and the tests requirements of MIL-PRF-49166 herein referred to in paragraphs 4.5.1.1, 4.5.1.4, 4.5.2, 4.5.4, 4.5.5, and 4.5.6.

4.3 Conformance inspection.

4.3.1 Tests. Perform the examination of 4.4 and the tests of 4.5 et el.

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4.4 Examination. Each Mechanical Scanner shall be examined for compliance with the requirements specified in 3.3 through 3.6. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection. The contractor/supplier may use his own test methods, measuring devices, tools, fixtures, and other equipment during the conduct of verification testing. This specification cites no test requirements as long as all the performance requirements per Section 3 are fully verified as being met. Tests shall be performed utilizing the Scan-Interlace , PL-1408/UAwith the MX9872(V)1/UA with a scan angle of 13.3, ± 0.5 degrees at the 60.0, ± 0.5 Hz scan rate.

4.5.1 Scan angle and frequency.

4.5.1.1 Variable scan frequency. See MIL-PRF-49166.

4.5.1.2 Specific scan frequency (60 Hz). The MX-9872(V)1/UA shall be connected to the Scan-Interlace, PL-1408/UA, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (Scan) transducer (Scan-Interlace, PL-1408/UA shall be monitored with a calibrated frequency counter. The Output of the Scan-Interlace, PL-1408/UA shall be adjusted to obtain the specified frequency. Not meeting the requirements of 3.5.1.2 constitutes failure.

4.5.1.3 Synchronous scan frequency. The MX-9872(V)1/UA shall be connected to Scan-Interlace, PL-1408/UA and mounted in a suitable fixture. Synchronization from an external stable source (such as a TV camera vertical sync pulse) and power shall be applied and the output of the Azimuth (scan) transducer (Scan-Interlace, PL-1408/UA) shall be monitored with a calibrated frequency counter. Not meeting the requirements of 3.5.1.3 constitutes failure.

4.5.1.4 Scan angle. See MIL-PRF-49166.

4.5.2 Interlace. See MIL-PRF-49166.

4.5.3 Scan direction. With the MX-9872(V)1/UA operating in either the 2:1 side or the 2:1 rear interlace mode, the outputs of the azimuth and interlace position transducers(Scan-Interlace, PL-1408/UA, shall be observed with an oscilloscope. The output waveforms shall show the applicable shape and phase relationships shown in Figure 1. Not meeting the requirements of 3.5.3 constitutes failure.

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4.5.4 Power consumption. See MIL-PRF-49166.

4.5.5 Scan efficiency. See MIL-PRF-49166.

4.5.6 Scan jitter in azimuth. See MIL-PRF-49166.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MX-9872(V)1/UA is intended for use with the AN/ASQ-170, Target Acquisition and Designation Sight, and AN/ASQ-11, Pilot Night Vision Sensor (TADS/PNVS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see5.1).

d. If First article tests are required.

1) Time frame for submission of first article test reports when first article tests are required.

2) Time frame for approval of first article test reports when first article tests are required.

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e. Production delivery schedule.

1) Defined in terms of monthly lots.

2) Actions required in case of Group C failures.

f. Marking requirements should be specified**.**

g. Refer to the General specification (MIL-PRF-49166) for other acquisition requirements data.

6.3 Definitions. (See MIL-PRF-49166)

6.4 Subject term (key word) listing.

Scan mechanism
Scan-Interlace

6.5 Changes form previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0113)

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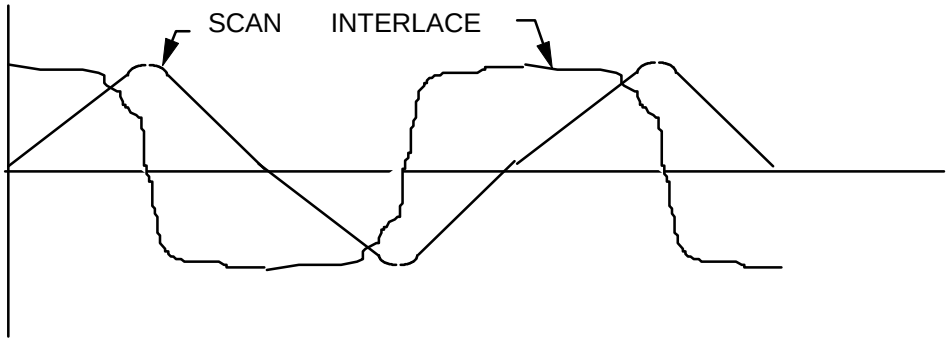
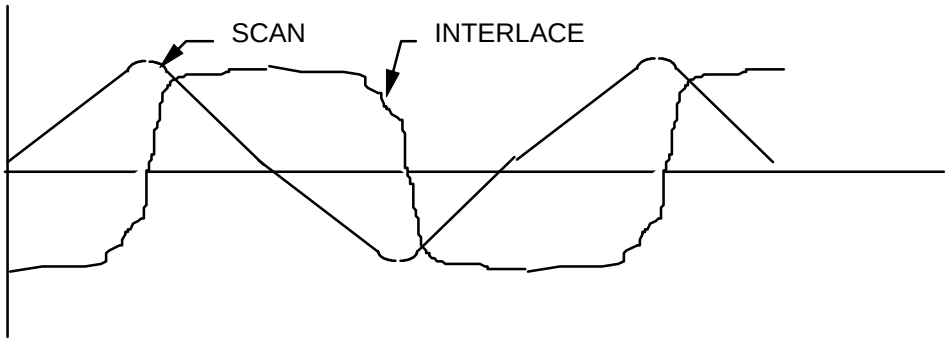
2:1 ‘REAR POSITION’ - INTERLACE LEADS SCAN APPROX 90°

PL-1408/UA
P1: CONNECT
PIN 33 to 34
PIN 17 to 4
Oscilloscope: (See Note)



2:1 ‘SIDE POSITION’ - INTERLACE FOLLOWS SCAN APPROX 90°

PL-1408/UA
P1: CONNECT
PIN 31 to 33
PIN 17 to 4
Oscilloscope: (See Note)



NOTE - CONNECT OSCILLOSCOPE AS FOLLOWS:

CHANNEL 1 - HIGH INPUT TO J3 (OR J2-7), LOW SCOPE TO GND.
CHANNEL 2 - HIGH INPUT TO J6 (OR J1-9), LOW SCOPE TO GND.

FIGURE 1. Interlace Transducer and Azimuth Position Transducer Outputs.

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